Bott Periodicity for the Unitary Group

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1 Introduction

The Bott periodicity theorem is a landmark in algebraic topology, establishing a surprising periodicity in the homotopy groups of the classical groups. While several modern approaches leverage the machinery of topological K-theory, clifford algebras, or the Atiyah–Singer index theorem, Bott's original proof [Bot59] strikes a geometric connection between the topology of Lie groups and the calculus of variations. The main realization is a Morse-theoretic description of the space of minimal geodesics.

For simplicity, we will focus on the case of complex Bott periodicity. Recall that a matrix $A \in \mathrm{GL}_n(\mathbb{C})$ is called unitary if $AA^\dagger = I$, where $A^\dagger$ denotes the conjugate transpose matrix of A. The unitary group $\mathrm{U}(n) \subset \mathrm{GL}_n(\mathbb{C})$ is the subgroup of unitary matrices. As the preimage of the identity under the smooth map

$$\begin{aligned} \mathrm{GL}_n(\mathbb{C}) &\longrightarrow \mathrm{GL}_n(\mathbb{C}) \\ A &\longmapsto AA^\dagger, \end{aligned}$$

$\mathrm{U}(n)$ is a closed subspace of $\mathrm{GL}_n(\mathbb{C})$, hence a Lie group [Ada69, Theorem 2.27]. As the columns of a unitary matrix are orthonormal vectors in $\mathbb{C}^n$, $\mathrm{U}(n)$ is bounded, hence compact by the Heine–Borel theorem.

1.1 The stable unitary group

For $m \leq n$, we have an inclusion

$$\begin{aligned} \iota_{mn} \colon \mathrm{U}(m) &\longrightarrow \mathrm{U}(n) \\ A &\longmapsto \begin{pmatrix} A & 0 \\ 0 & I_{n-m} \end{pmatrix}, \end{aligned}$$

where I_{n-m} is the $(n-m) \times (n-m)$ identity matrix. Indeed, $\iota_{mn}(A)$ is seen to be unitary from the calculation

$$\iota_{mn}(A)\iota_{mn}(A)^\dagger = \iota_{mn}(A)\iota_{mn}(A^\dagger) = \iota_{mn}(A)\iota_{mn}(A^{-1}) = \iota_{mn}(A)\iota_{mn}(A)^{-1} = I_m.$$

We define the stable unitary group U to be the direct limit $\varinjlim \mathrm{U}(n)$ of the directed system $\langle \mathrm{U}(n), \iota_{mn} \rangle$. Concretely,

$$\mathrm{U} = \bigsqcup_{n \in \mathbb{N}} \mathrm{U}(n)/\sim,$$

where $A_n \in \mathrm{U}(n) \sim A_m \in \mathrm{U}(m)$ if there exists $k \geq m, n$ such that

$$\iota_{nk}(A_n) = \begin{pmatrix} A_n & 0 \\ 0 & I_{k-n} \end{pmatrix} = \begin{pmatrix} A_m & 0 \\ 0 & I_{k-m} \end{pmatrix} = \iota_{mk}(A_m).$$

The complex Bott periodicity theorem states that $\pi_{i-1}\,\mathrm{U} \cong \pi_{i+1}\,\mathrm{U}$ for $i \geq 1$.

2 Homotopy Theory of the Unitary Group

2.1 Preliminaries on fibre bundles

Let G be a Lie group and M a smooth manifold. An action of the group G on the set M is said to be

(i) *smooth* if the map $G \times M \to M$ defined by $(g, m) \mapsto g \cdot m$ is smooth.

(ii) *free* if whenever $g \cdot m$ for some $m \in M$, we have $g = e$.

(iii) *proper* if the map $G \times M \to M \times M$ given by $(g, m) \mapsto (m, g \cdot m)$ is proper, in the sense that the preimage of a compact set is compact.

It is well-known [Lee12, Theorem 21.10] that a smooth, free, and proper action of a Lie group G on a smooth manifold M defines a unique smooth structure on the quotient space M/G such that the quotient map $q \colon M \to M/G$ is a submersion. If M is compact, q is automatically proper. Then Ehresmann's fibration theorem [Dun18, Theorem 8.5.10] states that

$$G \to M \to M/G$$

is a fibre bundle, in particular a Serre fibration [Swi02, Proposition 4.10]. It will be crucial to know that a Serre fibration

$$F \to E \to B$$

induces a long exact sequence of homotopy groups [Swi02, pages 55-56]:

$$\cdots \to \pi_{i+1}B \to \pi_i F \to \pi_i E \to \pi_i B \to \cdots.$$

2.2 Stability of homotopy groups

At this point, it is unclear why one should care about the homotopy groups of such an abstract object as U. Thus we make it our first goal to show that the homotopy groups $\pi_i \mathrm{U}(n)$ stabilize for large n, depending on i. More precisely:

Proposition 2.1. *If $i \leq 2n-1$, then for all $k \in \mathbb{N}$,*

$$\pi_i \mathrm{U}(n) \cong \pi_i \mathrm{U}(n+k).$$

Proof. $\mathrm{U}(n)$ acts smoothly, freely, and properly on $\mathrm{U}(n+1)$ by right multiplication, where we identify $\mathrm{U}(n)$ with its image under the inclusion $\iota_{n,n+1}\colon \mathrm{U}(n) \to \mathrm{U}(n+1)$. As above, this exhibits $\mathrm{U}(n+1)$ as a $\mathrm{U}(n)$-bundle over $\mathrm{U}(n+1)/\mathrm{U}(n)$. Since the cosets are parametrized by the last columns of matrices in $\mathrm{U}(n+1)$, which are unit vectors in $\mathbb{C}^{n+1}$, we identify the quotient with S^{2n+1}. The Serre fibration

$$\mathrm{U}(n) \to \mathrm{U}(n+1) \to S^{2n+1}$$

induces a long exact sequence of homotopy groups

$$\cdots \to \pi_{i+1} S^{2n+1} \to \pi_i \mathrm{U}(n) \to \pi_i \mathrm{U}(n+1) \to \pi_i S^{2n+1} \to \cdots .$$

Provided that $i \leq 2n-1$, we know that $\pi_{i+1} S^{2n+1} = 0$ and $\pi_i S^{2n+1} = 0$, so by exactness the map $\pi_i \mathrm{U}(n) \to \pi_i \mathrm{U}(n+1)$ induced by $\iota_{n,n+1}$ is an isomorphism. By induction, we get $\pi_i \mathrm{U}(n) \cong \pi_i \mathrm{U}(n+k)$ for all k. □

Remark 2.2. Since $\pi_{2n} S^{2n+1} = 0$, we see that $\pi_{2n} \mathrm{U}(n) \to \pi_{2n} \mathrm{U}(n+1)$ is surjective.

Since homotopy groups commute with filtered colimits for nice spaces [May99, page 67], we have

$$\pi_i \left(\varinjlim \mathrm{U}(n) \right) \cong \varinjlim \pi_i \mathrm{U}(n).$$

Hence the preceding proposition tells us that $\pi_i \mathrm{U}$ computes $\pi_i \mathrm{U}(n)$ for $n \geq \frac{i+1}{2}$.

2.3 Homotopy of the Stiefel and Grassmann manifolds

Recall that an orthonormal k-frame in $\mathbb{C}^m$ is an ordered set of k orthonormal vectors in $\mathbb{C}^m$. As in the previous proposition, the action of $\mathrm{U}(m-k)$ on $\mathrm{U}(m)$ for $k < m$ defines a fibre bundle

$$\mathrm{U}(m-k) \to \mathrm{U}(m) \to \mathrm{U}(m)/\mathrm{U}(m-k), \tag{2.1}$$

where an element of the quotient is identified with the last k columns of a matrix in $\mathrm{U}(m)$. Since the columns of a unitary matrix are orthonormal vectors in $\mathbb{C}^m$, we define the Stiefel manifold $V_k(\mathbb{C}^m) := \mathrm{U}(m)/\mathrm{U}(m-k)$ to be the space of orthonormal k-frames in $\mathbb{C}^m$.

Furthermore, $\mathrm{U}(k)$ acts on $V_k(\mathbb{C}^m)$ by multiplication on the right, where we view an k-frame in $\mathbb{C}^m$ as an $m \times k$ matrix. The orbits of this action consist of the k-frames in $\mathbb{C}^m$ which span the same linear subspace, so we define the Grassmann manifold $\mathrm{Gr}_k(\mathbb{C}^m) := \mathrm{U}(m)/\mathrm{U}(k) \times \mathrm{U}(m-k)$ to be the space of k-dimensional linear subspaces of $\mathbb{C}^m$. We use the long exact sequences associated to these fibre bundles to compute the homotopy groups of the Stiefel and Grassmann manifolds.

Proposition 2.3. *If $i \leq 2n$, then $\pi_i V_n(\mathbb{C}^{2n}) = 0$.*

Proof. The long exact sequence of the Serre fibration $\mathrm{U}(n) \to \mathrm{U}(2n) \to V_n(\mathbb{C}^{2n})$ is

$$\cdots \to \pi_i \,\mathrm{U}(n) \to \pi_i \,\mathrm{U}(2n) \to \pi_i V_n(\mathbb{C}^{2n}) \to \pi_{i-1} \,\mathrm{U}(n) \to \pi_{i-1} \,\mathrm{U}(2n) \to \cdots .$$

We know from Proposition 2.1 that $\pi_{i-1} \,\mathrm{U}(n) \to \pi_{i-1} \,\mathrm{U}(2n)$ is an isomorphism for $i \leq 2n$, so $\pi_i V_n(\mathbb{C}^{2n}) \to \pi_{i-1} \,\mathrm{U}(n)$ is the zero homomorphism. By exactness at $\pi_i V_n(\mathbb{C}^{2n})$, the map $\pi_i \,\mathrm{U}(2n) \to \pi_i V_n(\mathbb{C}^{2n})$ must be surjective. But Remark 2.2 implies this map is zero, showing that $\pi_i V_n(\mathbb{C}^{2n}) = 0$. □

Proposition 2.4. *If $i \leq 2n$, then $\pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \cong \pi_{i-1} \,\mathrm{U}(n)$.*

Proof. The long exact sequence of the fibre bundle $\mathrm{U}(n) \to V_n(\mathbb{C}^{2n}) \to \mathrm{Gr}_n(\mathbb{C}^{2n})$ is

$$\cdots \to \pi_i V_n(\mathbb{C}^{2n}) \to \pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \to \pi_{i-1} \,\mathrm{U}(n) \to \pi_{i-1} V_n(\mathbb{C}^{2n}) \to \cdots .$$

By Proposition 2.3, $\pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \to \pi_{i-1} \,\mathrm{U}(n)$ is an isomorphism. □

2.4 Homotopy of the special unitary group

Proposition 2.5. *If $i \geq 2$, then $\pi_i \,\mathrm{SU}(n) \cong \pi_i \,\mathrm{U}(n)$ for all $n \in \mathbb{N}$.*

Proof. Let $\mathrm{SU}(n)$ act on $\mathrm{U}(n)$ by multiplication.

The fibre bundle $\mathrm{SU}(n) \to \mathrm{U}(n) \to S^1$ induces the long exact sequence

$$\cdots \to \pi_{i+1} S^1 \to \pi_i \,\mathrm{SU}(n) \to \pi_i \,\mathrm{U}(n) \to \pi_i S^1 \to \cdots .$$

If $i \geq 2$ then $\pi_{i+1} S^1 = \pi_i S^1 = 0$, so $\pi_i \,\mathrm{SU}(n) \cong \pi_i \,\mathrm{U}(n)$. □

The next section will be devoted to showing that $\pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \cong \pi_{i+1} \,\mathrm{SU}(2n)$ for $i \leq 2n$. Then from the chain of isomorphisms

$$\pi_{i-1} \,\mathrm{U}(n) \cong \pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \cong \pi_{i+1} \,\mathrm{SU}(2n) \cong \pi_{i+1} \,\mathrm{U}(2n) \tag{2.2}$$

along with the earlier observation that $\pi_i \left(\varinjlim \mathrm{U}(n)\right) \cong \varinjlim \pi_i \,\mathrm{U}(n)$, we will deduce the Bott periodicity theorem for the unitary group:

Theorem 2.6 (Bott)**.** *For $i \geq 1$, $\pi_{i-1} \,\mathrm{U} \cong \pi_{i+1} \,\mathrm{U}$.*

3 Morse Theory for the Special Unitary Group

In this section we complete the proof of Bott periodicity for the unitary group by constructing isomorphisms

$$\pi_i \,\mathrm{Gr}_n(\mathbb{C}^{2n}) \cong \pi_{i+1} \,\mathrm{SU}(2n) \quad \text{for} \quad i \leq 2n.$$

Our main input will be the following theorem from Morse theory [Mil63, Corollary 22.2]:

Theorem 3.1. *Let M be a complete Riemannian manifold, and let $p, q \in M$ be two points with distance $\sqrt{d}$. Suppose the space Ω^d of minimal geodesics from p to q is a topological manifold, and every non-minimal geodesic from p to q has index $\geq \lambda_0$. Then*

$$\pi_i \Omega^d \cong \pi_{i+1} M \quad \textit{for} \quad 0 \leq i \leq \lambda_0 - 2.$$

3.1 Preliminaries on Riemannian geometry and Morse theory

We begin by defining various terms in the statement of Theorem 3.1.

Definition 3.2. A *Riemannian metric* g on a smooth manifold M is an inner product $g_p(-,-)\colon T_pM \times T_pM \to \mathbb{R}$ for each $p \in M$, which depends smoothly on p.

Formally, the smooth dependence means g is a smooth global section of the vector bundle $TM^* \otimes TM^*$. A smooth manifold with a Riemannian metric is called a *Riemannian manifold*.

Definition 3.3. If $\gamma\colon [0,1] \to M$ is a curve on a Riemannian manifold (M,g), then we define its *length* to be

$$\ell_g(\gamma) := \int_0^1 \sqrt{g_{\gamma(t)}(\gamma'(t),\gamma'(t))}\, dt.$$

We define the *distance* $d_g(p,q)$ between points $p,q \in M$ to be the infimum of $\ell_g(\gamma)$ over the set Ω_p^q of all curves γ with endpoints p and q. We leave it to the reader to verify that this makes M a metric space. A Riemannian manifold (M,g) is said to be *complete* if (M,d_g) is a complete metric space.

Definition 3.4. Let $\gamma\colon [0,1] \to M$ be a curve on a Riemannian manifold. We define its *energy* to be

$$E_g(\gamma) := \int_0^1 g_{\gamma(t)}(\gamma'(t),\gamma'(t))\, dt.$$

A *smooth deformation* of a curve $\gamma \in \Omega_p^q$ is a smooth map $\gamma\colon (-\epsilon,\epsilon) \times [0,1] \to M$, denoted by $\gamma_s(t) := \gamma(s,t)$, such that $\gamma_0(t) = \gamma(t)$ for $t \in [0,1]$ and $\gamma_s(0) = p$, $\gamma_s(1) = q$ for $s \in (-\epsilon,\epsilon)$. A curve $\gamma \in \Omega_p^q$ is called a *geodesic* if for all smooth deformations γ_s,

$$\left.\frac{d}{ds}\right|_{s=0} E_g(\gamma_s) = 0.$$

A geodesic $\gamma \in \Omega_p^q$ is called *minimal* if the infimum in the definition of $d_g(p,q)$ is attained at γ; otherwise γ is *non-minimal*.

Example 3.5. *A compact Lie group G admits a bi-invariant Riemannian metric by left-translating an* Ad*-invariant inner product on $\mathfrak{g} := T_eG$. In this case, the geodesics are simply left-translates of* 1*-parameter subgroups $t \mapsto \exp(tX)$, for $X \in \mathfrak{g}$.*

Example 3.6. *On* $\mathrm{SU}(2n)$*, we will use the inner product $\langle X,Y\rangle := \mathfrak{Re}(\mathrm{tr}(XY^\dagger))$ on $\mathfrak{su}_{2n}$. Recall that $\mathfrak{su}_{2n}$ consists of the skew-Hermitian $2n \times 2n$ matrices with trace* 0*. As a compact submanifold of $\mathbb{R}^{8n^2}$,* $\mathrm{SU}(2n)$ *is a complete Riemannian manifold. The remaining hypotheses of Theorem 3.1 will be more difficult to satisfy.*

The remainder of this section relies on the construction of the Riemann curvature tensor [Mil63, Section 9]. Although we provide general statements, we will skip the gory details and we simply note that in the case of a Lie group with a bi-invariant Riemannian metric, the Riemann curvature tensor is given on left-invariant vector fields X,Y,Z by

$$\mathcal{R}(X,Y)Z = \frac{1}{4}[[X,Y],Z].$$

Definition 3.7. Let $\gamma\colon [0,1] \to M$ be a geodesic and $V = \frac{d\gamma}{dt}$. A vector field J along γ is called a *Jacobi field* if it satisfies the Jacobi differential equation

$$\frac{d^2 J}{dt^2} + \mathcal{R}(V, J)V = 0.$$

Two distinct points $p = \gamma(a)$ and $q = \gamma(b)$ are called *conjugate* along γ with multiplicity m if there exist m linearly independent Jacobi fields along γ which vanish for $t = a, b$.

Definition 3.8. The *index* of a geodesic $\gamma\colon [0,1] \to M$ is equal to the number of points $\gamma(t)$, for $t \in (0,1)$, conjugate to $\gamma(0)$ along γ, counted with multiplicity.

Remark 3.9. This definition is non-standard, although it will simplify our present work. In general, we define the index of any nondegenerate critical point of a smooth real-valued function to be the number of negative eigenvalues of its Hessian matrix, counted with multiplicity. We can then specialize to the case of the energy function, whose critical points are precisely the geodesics. The equivalence with our definition is the content of the Morse Index Theorem [Mil63, Theorem 15.1].

3.2 Minimal geodesics on the special unitary group

To apply Morse theory to $\mathrm{SU}(2n)$, we study the space $\Omega^{\pi\sqrt{2n}}$ of minimal geodesics on $\mathrm{SU}(2n)$ from I to $-I$, endowed with the compact-open topology. As per Theorem 3.1, we would like it to be a manifold. Indeed, we will identify $\Omega^{\pi\sqrt{2n}}$ with the Grassmannian $\mathrm{Gr}_n(\mathbb{C}^{2n})$. Consider

$$D = \begin{pmatrix} i\pi I_n & 0 \\ 0 & -i\pi I_n \end{pmatrix} \in \mathrm{SU}(2n).$$

We claim that the map

$$\begin{aligned} f\colon \mathrm{U}(2n) &\longrightarrow \Omega^{\pi\sqrt{2n}} \\ A &\longmapsto \left(t \mapsto \exp\left(tADA^\dagger\right)\right) \end{aligned}$$

descends to a homeomorphism $\overline{f}\colon \mathrm{Gr}_n(\mathbb{C}^{2n}) \to \Omega^{\pi\sqrt{2n}}$. Indeed if

$$B = \begin{pmatrix} B_1 & 0 \\ 0 & B_2 \end{pmatrix} \in \mathrm{U}(n) \times \mathrm{U}(n) \subset \mathrm{U}(2n),$$

then $BDB^\dagger = D$. Then for any $A \in \mathrm{U}(2n)$, we have

$$(AB)D(AB)^\dagger = A(BDB^\dagger)A^\dagger = ADA^\dagger,$$

showing that f descends to a well-defined map on $\mathrm{Gr}_n(\mathbb{C}^{2n})$. We remark that $\mathrm{Gr}_n(\mathbb{C}^{2n})$ is compact, as a quotient of the compact Lie group $\mathrm{U}(2n)$, and $\Omega^{\pi\sqrt{2n}}$ is Hausdorff, as a subspace of the path space of a Hausdorff space. Thus to show that $\overline{f}$ is a homeomorphism, it suffices to show it is a continuous bijection. It is continuous in the compact-open topology because the adjoint map

$$\begin{aligned} \mathrm{Gr}_n(\mathbb{C}^{2n}) \times [0,1] &\longrightarrow \mathrm{SU}(2n) \\ ([A], t) &\longmapsto \exp\left(tADA^\dagger\right) \end{aligned}$$

is visibly continuous. To see $\overline{f}$ is surjective, we know that a geodesic $\gamma\colon [0,1] \to \mathrm{SU}(2n)$ starting at I takes the form $\exp(tX)$ for $X \in \mathfrak{su}_{2n}$. Since X is skew-Hermitian, there exists $A \in \mathrm{U}(2n)$ such that $A^{-1}XA$ is diagonal. Moreover the condition $\gamma(1) = \exp(X) = -I$ implies

$$\exp\big(A^{-1}XA\big) = A^{-1}\exp(X)A = A^{-1}(-I)A = -I,$$

hence the entries of $A^{-1}XA$ must be of the form $k\pi i$ for odd integers k. Since the length of γ is

$$\sqrt{\mathrm{tr}(XX^\dagger)} = \sqrt{\mathrm{tr}\Big((A^{-1}XA)(A^{-1}XA)^\dagger\Big)} = \pi\sqrt{k_1^2 + \cdots + k_{2n}^2},$$

we see that X determines a minimal geodesic if and only if $k_j = \pm 1$ for all $j = 1, \ldots, 2n$. But X is traceless, so 1 and -1 must each appear exactly n times, and thus by permuting the diagonal entries we may assume $A^{-1}XA = D$. Therefore

$$\overline{f}([A]) = \exp\Big(tADA^\dagger\Big) = \exp(tX) = \gamma(t),$$

so $\overline{f}$ is surjective.

Finally if $\overline{f}([A]) = \overline{f}([B])$, then by differentiating at $t = 0$ we get $ADA^\dagger = BDB^\dagger$. Rearranging, we see that

$$(B^\dagger A)D = D(B^\dagger A),$$

and some linear algebra shows that $B^\dagger A$ must have the same block diagonal form as D, and thus lies in $U(n) \times U(n)$. We conclude that $[A] = [B]$, so $\overline{f}\colon \mathrm{Gr}_n(\mathbb{C}^{2n}) \to \Omega^{\pi\sqrt{2n}}$ is a homeomorphism.

3.3 The index of non-minimal geodesics

The final hypothesis of Theorem 3.1 concerns the index of non-minimal geodesics. We first illustrate a method for calculating the index of geodesics in a Lie group:

Lemma 3.10. *Let $\gamma\colon [0,1] \to G$ be a geodesic in a Lie group with bi-invariant Riemannian metric. Let $V = \frac{d\gamma}{dt}\Big|_{t=0}$ be the velocity vector at $p = \gamma(0)$. Define a linear transformation*

$$\begin{aligned} K_V\colon T_pM &\longrightarrow T_pM \\ W &\longmapsto \mathcal{R}(V,W)V. \end{aligned}$$

Then the conjugate points to p along γ are given by $\gamma\left(\frac{k\pi}{\sqrt{\mu}}\right)$ for nonzero integers k and positive eigenvalues μ of K_V, with multiplicity.

Proof sketch. Recall that for Lie groups,

$$K_V(W) = \mathcal{R}(V,W)V = [[V,W],V] = -[V,[V,W]] = -\frac{1}{4}\,\mathrm{ad}_V^2(W).$$

In particular since $\langle\cdot,\cdot\rangle$ is Ad-invariant, K_V is self-adjoint:

$$\langle K_V(W), W'\rangle = -\frac{1}{4}\langle \mathrm{ad}_V^2(W), W'\rangle = -\frac{1}{4}\langle W, \mathrm{ad}_V^2(W')\rangle = \langle W, K_V(W')\rangle.$$

Pick an orthonormal basis $X_1, \ldots, X_r$ of T_pM consisting of eigenvectors for K_V:

$$K_V(X_i) = \mu_i X_i \quad \text{for} \quad i = 1, \ldots, r.$$

By parallel transport, we may extend the X_i to linearly independent vector fields $\mathfrak{X}_i$ along γ. By certain symmetry properties of Lie groups, the curvature tensor is invariant under parallel transport, so we have

$$K_V(\mathfrak{X}_i(\gamma(t))) = \mu_i \mathfrak{X}_i(\gamma(t)) \quad \text{for} \quad t \in [0,1].$$

We wish to find nonzero Jacobi fields vanishing at $t = 0$. Writing $J(t) = \sum_{i=1}^r j_i(t)\mathfrak{X}_i(\gamma(t))$, the Jacobi equation becomes

$$0 = \sum_{i=1}^r \frac{d^2 j_i}{dt^2}\mathfrak{X}_i + K_V(\mathfrak{X}_i) = \sum_{i=1}^r \frac{d^2 j_i}{dt^2}\mathfrak{X}_i + \mu_i j_i \mathfrak{X}_i.$$

Since the $\mathfrak{X}_i$ are linearly independent, we obtain the system

$$\begin{cases} \frac{d^2 j_i}{dt^2} + \mu_i j_i = 0, \\ j_i(0) = 0. \end{cases}$$

If $\mu_i > 0$, we have the solutions $j_i(t) = c_i \sin\left(\sqrt{\mu_i} t\right)$ for constants c_i. Its zeroes are multiples of $\frac{\pi}{\sqrt{\mu_i}}$. If $\mu_i = 0$ then $j_i(t) = c_i t$, and if $\mu_i < 0$, then $j_i(t) = c_i \sinh\left(\sqrt{|\mu_i|} t\right)$. Thus there exist zeroes distinct from $t = 0$ only if $\mu_i > 0$, and in this case the conjugate points are given by $\gamma\left(\frac{k\pi}{\sqrt{\mu}}\right)$ for nonzero integers k and positive eigenvalues μ. □

Lemma 3.11. *Every non-minimal geodesic from I to $-I$ in* $\mathrm{SU}(2n)$ *has index* $\geq 2n + 2$.

Proof. Of course we will use Lemma 3.10. Recall a geodesic from I to $-I$ takes the form $t \mapsto \exp(tX)$ for a skew-Hermitian $2n \times 2n$ matrix $X \in \mathfrak{su}_{2n}$, having trace zero and eigenvalues $k_j \pi i$ for odd integers k_j with sum zero. We may assume without loss of generality that A is diagonal with the k_j decreasing from left to right. If $X = (x_{ij}) \in \mathfrak{su}_{2n}$, then we compute

$$\begin{aligned} [A, X] &= (i\pi(k_j - k_\ell)x_{j\ell}), \\ [A, [A, X]] &= (-\pi^2(k_j - k_\ell)x_{j\ell}), \\ K_A(X) &= \left(\frac{\pi^2}{4}(k_j - k_\ell)^2 x_{j\ell}\right). \end{aligned}$$

We can then find a basis for $\mathfrak{su}_{2n}$ consisting of eigenvectors of K_A, namely:

(i) for $j < \ell$, the matrix $E_{j\ell} - E_{\ell j}$ with eigenvalue $\frac{\pi^2}{4}(k_j - k_\ell)^2$,

(ii) for $j < \ell$, the matrix $iE_{j\ell} + iE_{\ell j}$ with eigenvalue $\frac{\pi^2}{4}(k_j - k_\ell)^2$,

(iii) the diagonal matrices in $\mathfrak{su}_{2n}$ with eigenvalue 0.

These are visibly linearly independent over $\mathbb{R}$. There are $\binom{2n}{2}$ matrices in (i), and symmetrically (ii), along with $2n-1$ diagonal matrices in $\mathfrak{su}_{2n}$. In total, we count

$$2 \cdot \binom{2n}{2} + 2n - 1 = 2n(2n-1) + 2n - 1 = 4n^2 - 1 = \dim_{\mathbb{R}} \mathfrak{su}_{2n},$$

so this is indeed a basis. We see that the positive eigenvalues of K_A are $\frac{\pi^2}{4}(k_j - k_\ell)^2$ with $k_j > k_\ell$, each having multiplicity two because of (i) and (ii). The multiples of $\frac{\pi}{\sqrt{\mu}}$ take the form

$$t = \frac{2}{k_j - k_\ell}, \frac{4}{k_j - k_\ell}, \frac{6}{k_j - k_\ell}, \dots.$$

In the open interval $(0,1)$, there are $\frac{k_j - k_\ell}{2} - 1$ such values. Therefore the index of γ is

$$\lambda = \sum_{k_j > k_\ell} k_j - k_\ell - 2.$$

Note that every term in this sum is nonnegative because the k_j are odd. Now suppose γ is a non-minimal geodesic. If at least $n+1$ of the k_j are negative, then because the k_j are odd and sum to zero, there must exist some $k_j \geq 3$, so by pairing with all the negative k_j, which are ≤ -1, we see that

$$\lambda \geq \sum_{1}^{n+1} (3 - (-1) - 2) = 2n + 2.$$

The case where at least $n+1$ are positive is similar. Finally if n of the k_j are negative and n are positive, then by non-minimality they are not all ± 1. In particular one must be ≥ 3 and another must be ≤ -3, so that

$$\lambda \geq (3 - (-3) - 2) + \sum_{1}^{n-1} (3 - (-1) - 2) + \sum_{1}^{n-1} (1 - (-3) - 2) = 4n \geq 2n + 2.$$

In any case, a non-minimal geodesic has index $\geq 2n+2$. □

3.4 Concluding remarks

We have thus shown $M = \mathrm{SU}(2n)$ satisfies the hypotheses of Theorem 3.1, and since $\Omega^{\pi\sqrt{2n}} \cong \mathrm{Gr}_n(\mathbb{C}^{2n})$, we obtain the desired isomorphism

$$\pi_i \, \mathrm{Gr}_n(\mathbb{C}^{2n}) \cong \pi_{i+1} \, \mathrm{SU}(2n) \quad \text{for} \quad i \leq 2n,$$

completing the chain of isomorphisms equation (2.2) and the proof of the Bott periodicity theorem. In fact, the explicit computations of $\pi_i \, \mathrm{U}$ follow from identifications $\mathrm{U}(1) \cong S^1$ and $\mathrm{SU}(2) \cong S^3$. It is well-known that

$$\pi_1(S^1) \cong \mathbb{Z} \quad \text{and} \quad \pi_2(S^3) \cong 0.$$

Since we saw in Proposition 2.1 that $\pi_i \, \mathrm{U} \cong \pi_i \, \mathrm{U}(n)$ for $i \leq 2n-1$, and in Proposition 2.5 that $\pi_i \, \mathrm{U}(n) \cong \pi_i \, \mathrm{SU}(n)$ for $i \geq 2$, we conclude that

$$\pi_1 \, \mathrm{U} \cong \pi_1 \, \mathrm{U}(1) \cong \mathbb{Z} \quad \text{and} \quad \pi_2 \, \mathrm{U} \cong \pi_2 \, \mathrm{U}(2) \cong \pi_2 \, \mathrm{SU}(2) \cong 0.$$

By periodicity, we obtain:

Corollary 3.12. *For $i \geq 0$, $\pi_{2i} \, \mathrm{U} \cong 0$ and $\pi_{2i+1} \, \mathrm{U} \cong \mathbb{Z}$.*

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